Copyright © 2003. Elsevier Science & Technology. All rights reserved.

C - 4

Perfect Maps

Dennis K. Burke, Oxford, OH, USA

A **perfect map** $f : X \to Y$ is a closed, continuous and onto map with $f^{-1}(y)$ ***compact*** in X for every $y \in Y$. In this article the term map (or mapping) will always mean a continuous map. For convenience, we assume all topological spaces considered here are ***Hausdorff spaces*** (even though for emphasis, Hausdorff may be indicated). By a **closed map** we mean that $f(A)$ is closed in Y for every closed set $A \subseteq X$. The point inverses $f^{-1}(y)$, for $y \in Y$, are sometimes called the **fibers** of the map f. It is sometimes interesting to consider a continuous closed map with a condition on the fibers which is weaker than compact (such as ***Lindelöf***) or with no condition at all. A closed map with ***countably compact*** fibers is called a **quasi-perfect map**. Many proofs of results quoted here may be found in [E] and a survey on closed maps and perfect maps may be found in [3].

A closed map $f : X \to Y$ is always a ***quotient*** map [E, Section 2.4]. If f is a quotient map it is possible to check conditions on the fibers of f to see if f is closed. If $W \subseteq X$ let the **saturated part** of W be given by the union of all fibers which are fully contained in W. This is the set $S(W) = f^{-1}(Y \setminus f(X \setminus W))$. A set $A \subseteq X$ would be called **saturated** (wrt f) if $A = S(A)$. We see that a quotient map $f : X \to Y$ is closed if and only if, for every open set $W \subseteq X$, the saturated part $S(W)$ is open. In the language of ***decomposition spaces*** (or ***quotient spaces***) this says that a decomposition space is ***upper semi-continuous*** if and only if the associated quotient map is closed. To construct a perfect map via the upper semi-continuous decomposition process it would be necessary to start with a partition $\mathbb{P}$ of X into compact subsets.

If $f : X \to Y$ is a closed map and $A \subseteq X$ is a saturated subset of X then $f|_A : A \to f(A)$ is a closed map and hence a quotient map − so closed maps are ***hereditarily quotient*** [E, 2.4.F]. However, the restriction of f to an arbitrary subset of X need not give a closed map or even a quotient map. If f is restricted to a closed subset $Z \subseteq X$ then $f|_Z : Z \to f(Z)$ is a closed map. Certainly, if f was actually a perfect map then $f|_Z$ would be a perfect map. The composition of two closed maps clearly yields a closed map. The composition of two perfect maps also gives a perfect map. This follows because, under any perfect map $f : X \to Y$, if $C \subseteq Y$ is a compact subset of Y then the preimage $f^{-1}(C)$ is a compact subset of X [E, 3.7.2]. This property of the inverse preservation of compactness under perfect maps is close to a characterization of perfect maps and is so when the range of the map is a Hausdorff *k*-***space***. That is, if $f : X \to Y$ is a continuous map, with the preimage $f^{-1}(C)$ compact in X for every compact $C \subseteq Y$ (sometimes called a **proper map**), and Y is a Hausdorff k-space then f is a perfect map.

Since compactness is preserved under a continuous map and compact subspaces of a Hausdorff space are closed, it is clear that any continuous map from a compact space X onto a Hausdorff space Y is a perfect map. If X is compact Hausdorff and Y is any space then the projection map $\pi_2 : X \times Y \to Y$ is a closed map [E,

Hart, K. P., et al. Encyclopedia of General Topology, Elsevier Science & Technology, 2003. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/willm/detail.action?docID=299054.
Created from willm on 2026-09-25 13:03:12.

Copyright © 2003. Elsevier Science & Technology. All rights reserved.

3.1.16] with fibers all homeomorphic to X. So π_2 is a perfect map. If X is an arbitrary Hausdorff space and, for all spaces Y, the projections $\pi_2 : X \times Y \to Y$, are closed maps then X, in fact, must be compact. If Z is a closed subspace of $X \times Y$ and X is compact then the restricted map $\pi_{2|Z}$ is a perfect map onto its range $\pi_{2|Z}$ (Z). In a sense, all perfect maps can be considered to arise in this fashion since it is known that a continuous onto map $g : Z \to Y$ (for ***Tychonoff space*** Z) is perfect if and only if for any ***compactification*** X of Z, the map $\# : Z \to X \times Y : \#(z) = (z, g(z))$ is an embedding map onto a closed subspace Z' of $X \times Y$. This fact is also the basis for the following useful observation: If P is a topological property which is closed hereditary then P is inversely preserved under perfect maps (with Tychonoff domain) if and only if $X \times Y$ has property P whenever X is compact and Y has property P.

When maps $f_1 : X_1 \to Y_1$ and $f_2 : X_2 \to Y_2$ are given it is natural to consider the **product map** $f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2$ defined in the obvious manner. If $f_1 \times f_2$ is a closed map then both f_1 and f_2 are closed. However, if f_1 is perfect and f_2 is closed the product map $f_1 \times f_2$ may not be closed [E, 2.3.28]. To illustrate by an example, let $id_{\mathbb{R}}$ be the identity map of the real line $\mathbb{R}$ onto itself (a nice perfect map) and let f_2 be the map of $\mathbb{R}$ onto the one-point discrete space {0}. The product map $id_{\mathbb{R}} \times f_2 : \mathbb{R} \times \mathbb{R} \to \mathbb{R} \times \{0\}$ is not closed. The graph $G = \{(x, y) : y = 1/x\}$ is a closed subset of $\mathbb{R} \times \mathbb{R}$, but the image of G under $id_{\mathbb{R}} \times f_2$ is not closed in $\mathbb{R} \times \{0\}$. If f_1 and f_2 are both perfect then $f_1 \times f_2$ is a perfect map (and conversely). A continuous map $f : X \to Y$ (with X Hausdorff) is perfect if and only if, for every Hausdorff space Z, the product map $f \times id_Z : X \times Z \to Y \times Z$ is closed [E, 3.7.14].

The above result about the product of two perfect maps can be upgraded to an infinite family $\{f_\alpha : \alpha \in \Lambda\}$ of (non-empty) maps. That is, the product map $\Pi_{\alpha\in\Lambda} f_\alpha$ is perfect if and only if, for every $\alpha \in \Lambda$, f_α is perfect. When all of the maps $f_\alpha : X \to Y_\alpha$ have a common domain X then we can talk about the **diagonal map** $\Delta_{\alpha\in\Lambda} f_\alpha : X \to \Pi_{\alpha\in\Lambda} Y_\alpha$, defined by $\Delta_{\alpha\in\Lambda} f_\alpha (x) = (f_\alpha(x))_{\alpha\in\Lambda}$. Since the ***diagonal*** Δ of $\Pi_{\alpha\in\Lambda} X$ is homeomorphic to X this carries the same information as the product map $\Pi_{\alpha\in\Lambda} f_\alpha$ restricted to the diagonal of $\Pi_{\alpha\in\Lambda} X$. If at least one f_β is perfect and all Y_α are Hausdorff then this is enough to say that the diagonal map $\Delta_{\alpha\in\Lambda} f_\alpha$ is perfect [E, 3.7.11].

Product maps are naturally involved in the study of ***inverse limit*** systems. Suppose $\mathbf{S} = \#X_n, f_n\#$ is an inverse limit sequence with $Y = \varprojlim X_n$. If all ***bonding maps*** f_n are perfect then all projection maps $\pi_{k|X} : \mathbf{X} \to X_k$ are perfect. (Here, π_k represents the usual projection map from $\Pi_{n\in\omega} X_n$ to X_k.) If $\mathbf{T} = \#Y_n, g_n\#$ is another inverse limit sequence with $X = \varprojlim X_n$ is a ***system map*** from $\mathbf{S}$ to $\mathbf{T}$, with all $\#_n$ perfect, then the induced map $\varphi = \varprojlim \varphi_n = \Pi_{n\in\omega} \varphi_n \mid X$ is a perfect map from $\mathbf{X}$ to $\mathbf{Y}$. Similar results hold for more general inverse systems [E].

A perfect map $f : X \to Y$ has the curious property that if f is extended continuously over a Hausdorff space $Z \supseteq X$ (with the same range Y) then X must be closed in Z [E, 3.7.4]. That is, X cannot be a dense proper subspace of Z. For Tychonoff spaces X, Y this gives the frequently useful result: A continuous map $f : X \to Y$ is perfect if and only if for any compactification bY of Y, the continuous extension $F : \beta X \to bY$ of f satisfies the condition that $F(\beta X \setminus X) \subseteq bY \setminus Y$. We also see that if X, Y are ***locally compact*** Hausdorff spaces and ωX, ωY

Hart, K. P., et al. Encyclopedia of General Topology, Elsevier Science & Technology, 2003. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/willm/detail.action?docID=299054.
Created from willm on 2026-09-25 13:03:12.

Copyright © 2003. Elsevier Science & Technology. All rights reserved.

denote the respective ***one-point compactifications*** then a map $f : X \to Y$ is perfect if and only if the extension of f obtained by assigning the point at infinity of ωX to the point at infinity of ωY gives a continuous extension.

Much of the utility of closed or perfect maps comes from results about the preservation of topological properties in the image or preimage direction. Many of the common "base axiom" properties are preserved in the image direction under perfect maps. This includes the properties of being ***metrizable, second-countable*** (more generally, with ***weight*** $\leqslant \kappa$), ***developable, quasi-developable, quasi-metrizable***, a γ-***space, stratifiable*** (for closed maps) or ***semistratifiable*** (closed maps), or having a σ-***locally countable*** base, ***uniform base***, σ-***point-finite*** base, ***point-countable*** base, $\delta\theta$-***base, base of countable order*** or ***primitive base***. Notable exceptions to this would include spaces with a σ-***disjoint*** base, ***first-countable*** spaces, ***semi-metrizable*** spaces, ***symmetrizable*** spaces and ***Nagata spaces*** − these classes are not generally preserved under perfect maps. It is interesting that the class of stratifiable spaces (same as $\boldsymbol{M_3}$***-spaces***) is even preserved under closed maps but it is not known whether the class of $\boldsymbol{M_1}$***-spaces*** is preserved under perfect maps. This question of whether M_1-spaces are preserved under perfect maps is actually equivalent to the well-known "$M_3 \Rightarrow M_1$" question [KV, Chapter 10]. However, M. Ito [HvM, Chapter 7] has shown that the class of hereditarily M_1-spaces is preserved under closed maps, and C. Borges and D.J. Lutzer have shown that M_1-spaces are preserved under perfect irreducible maps.

The preservation result for metrizable spaces is worthy of special note. The **Hanai–Morita–Stone Theorem** [E, 4.4.17] says that if $f : X \to Y$ is a closed continuous map from a ***metric*** space X onto a space Y then Y is metrizable $\iff$ Y is first-countable $\iff$ for every $y \in Y$, $\partial f^{-1}(y)$ is compact. Not only do we see that perfect maps preserve metrizability (and complete metrizability) but if a closed continuous map f does preserve metrizability then f must have fibers with compact boundaries. In this sense f is close to being a perfect map. Indeed, for this f, there is always a closed subspace Z of X such that $f|_Z$ is a perfect map from Z onto Y. Such a map is said to be **inductively perfect**. In case X is connected (so every fiber has a non-empty boundary) the closed set Z can simply be $Z = X \setminus \cup\{\text{Int } f^{-1}(y): y \in Y\}$. A stronger result in this direction is given by E.A. Michael [10]. He shows that if X is ***paracompact***, Y is first-countable or ***locally compact*** and $f : X \to Y$ is a closed continuous map then $\partial f^{-1}(y)$ is compact for every $y \in Y$; hence f is inductively perfect. An interesting corollary to this says that any closed map f from a paracompact space X onto any space Y is compact-covering. (A map $f : X \to Y$ is said to be **compact-covering** if every compact subset of Y is the image of some compact subset of X under f.) A classic result by Michael says that an open map f from metric X onto paracompact Y such that all fibers are complete wrt the metric on X, is inductively perfect. There are several variations and improvements on this result − see [vMR, Chapter 17] for a discussion and a beginning on references.

Since perfect maps are compact-covering it is clear that inductively perfect maps are also compact-covering maps. It is natural to ask when compact-covering maps are inductively perfect [vMR, Chapter 17]. Answering a question by Michael, G. Debs and J. Saint Raymond have given an example of a compact-covering map $f : X \to Y$ between ***separable*** metric spaces X, Y such that every fiber of f is compact but f is not inductively perfect. In [5], Debs and Saint Raymond have continued a descriptive set-theoretic approach to the validity of the statement $\mathbb{A}(X, Y)$, for separable metric spaces X and Y. $\mathbb{A}(X, Y)$: "Any compact-covering map $f : X \to Y$ is inductively perfect". It is known (J.P.R. Christensen; J. Saint Raymond) that $\mathbb{A}(X, Y)$ holds in **ZFC** if X is a

Hart, K. P., et al. Encyclopedia of General Topology, Elsevier Science & Technology, 2003. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/willm/detail.action?docID=299054.
Created from willm on 2026-09-25 13:03:12.

Copyright © 2003. Elsevier Science & Technology. All rights reserved.

Polish space (= separable complete metrizable). A.V. Ostrovskiĭ has shown that $\mathcal{A}(X, Y)$ holds in ZFC if Y is σ-***compact***. Debs and Saint Raymond have shown that in the universe L, there exists an F_σ-subset X of ω^ω and a compact-covering map $f : X \to \omega^\omega$ where f is not inductively perfect. See [5] for references.

Closed maps preserve several important covering properties in the image direction. The classes of paracompact spaces, ***metacompact*** spaces, ***subparacompact*** spaces, and ***submetacompact*** spaces are all preserved under closed map images. The classes of ***para-Lindelöf*** and σ-para-Lindelöf spaces are preserved under perfect images. See [KV, Chapter 9] for a more complete list with corresponding proofs. The classes of ***screenable*** spaces and ***orthocompact*** spaces are not generally preserved under closed maps or even perfect maps, and ***countably paracompact*** spaces are not preserved under closed maps. It is not known whether the class of ***meta-Lindelöf*** spaces is preserved under perfect maps. All of the covering properties mentioned in this paragraph, except orthocompactness, are inversely preserved under closed maps with Lindelöf fibers (and ***regular*** domain).

When showing that topological properties are preserved under a closed or perfect map it is often useful to rely on certain structures of collections of sets being preserved in the image direction. If $f : X \to Y$ is closed and $\mathcal{A}$ is a ***closure-preserving*** collection in X then $\{f(A): A \in \mathcal{A}\}$ is closure-preserving in Y. If f is perfect and $\mathcal{A}$ is ***locally finite*** in X then $\{f(A): A \in \mathcal{A}\}$ is locally finite in Y. If $\mathcal{A}$ is a σ-***relatively-discrete*** cover of X (and f is perfect) then Y has a σ-relatively-discrete cover ε which is a refinement of $\mathcal{A}$ [4]. Moreover, if $\mathcal{A}$ is a ***network*** for X then ε can be chosen to be a network for Y.

The phrase **perfect class** of spaces [E] is used to describe a class $\mathbb{C}$ of Hausdorff topological spaces which is closed under the formation of perfect images and perfect preimages of spaces from $\mathbb{C}$. The class of compact spaces is certainly a well-known perfect class. Other examples of perfect classes of spaces include the classes of regular spaces, Lindelöf spaces, paracompact spaces, σ-compact spaces, countably compact spaces, locally compact spaces, ***Čech-complete*** spaces (restricted to Tychonoff spaces), spaces of ***countable type, strict p-space***s, Σ-***space***s and k-spaces. The class of ***normal*** spaces is not perfect − normality is certainly preserved under a continuous closed map but for the failure of the inverse image direction, look at the space $(\omega_1 + 1) \times \omega_1$ [E, 3.12.20], where $(\omega_1 + 1)$ and ω_1 have the ***order topology***. This is a well-known non-normal countably compact space and the projection π_2: $(\omega_1 + 1) \times \omega_1 \to \omega_1$ is a perfect map onto the normal space ω_1. The Tychonoff property is not generally preserved in either the image or preimage direction.

Other classes known to be invariant under closed images include ***sequential*** (quotient maps), normal, ***realcompact, Fréchet space***s, ***monotonically normal*** spaces, ***collectionwise normal*** spaces, σ-***space***s and spaces with all closed sets G_δ. The classes of ***scattered*** spaces, ***pseudocompact*** spaces, ***Čech-analytic*** [HvM, Chapter 8] (with Tychonoff range) [8], and countably paracompact spaces are preserved under perfect images. Regular spaces with ***property D*** are preserved under finite-to-one perfect maps [15]. Spaces of ***point-countable type, p-spaces*** [6], ***q-spaces***, spaces with a ***G_δ-diagonal, Baire spaces, submetrizable*** spaces, realcompact spaces, ***elastic spaces, zero-dimensional*** spaces, ***extremally disconnected*** spaces and spaces with ***property wD*** [15] are not generally preserved to the image under a perfect map. The property of being a G_δ-set is not preserved under a perfect map. It is not known if the class of ***wΔ-space***s is preserved under perfect maps.

A topological property will not be inversely preserved under a perfect map unless compact spaces always have that property. Properties which are inversely preserved under perfect maps (and not yet mentioned

Hart, K. P., et al. Encyclopedia of General Topology, Elsevier Science & Technology, 2003. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/willm/detail.action?docID=299054.
Created from willm on 2026-09-25 13:03:12.

Copyright © 2003. Elsevier Science & Technology. All rights reserved.

above) include point-countable type, q-spaces, realcompact, regular property D (quasi-perfect) and property wD (quasi-perfect).

There is a general situation when certain topological properties can be "pulled back" under a perfect map [E, 3.7.27]: Suppose P is a closed hereditary topological property which is also finitely productive. If there exists a perfect map $f : X \to Y$ where Y has property P and there exists a **condensation** (one-to-one continuous map) $g : X \to Z$ onto a space Z with property P then X also has property P. The map g tends to give X and fibers of f certain properties which may otherwise not be present. For example, if Z has a G_δ-diagonal then so does X and all fibers of f would be metrizable. This result, due to A. Arhangel'skiĭ has been strengthened (in some cases) by replacing the condensation space Z with a cleavability condition. If $\mathbb{P}$ is a class of spaces, a space X is said to be **cleavable** over $\mathbb{P}$ if for every $A \subseteq X$ there is some $Z \in \mathbb{P}$ and a map $g : X \to Z$ with $A = f^{-1}(f(A))$. Clearly, if X admits a condensation onto some $Z \in \mathbb{P}$ then X is cleavable over $\mathbb{P}$. See [HvM, Chapter 7] for references to the following: Let $\mathbb{P}$ be one of the classes of metric spaces, ***Moore spaces***, σ-spaces, ***semi-metric*** spaces, (semi)-stratifiable spaces or spaces with a G_δ-diagonal. If X admits a perfect map onto an element of $\mathbb{P}$ and is cleavable over $\mathbb{P}$ then $X \in \mathbb{P}$.

The class of Baire spaces deserves special attention. As mentioned above, Baire spaces are not generally preserved under perfect maps. However, Baire spaces are preserved and inversely preserved under closed irreducible maps [1]. A map $f : X \to Y$ is said to be **irreducible** if the only closed subset Z of X, with $f(Z) = Y$, is $Z = X$. Equivalently, if W is any non-empty open subset of X then the saturated part S $(W) \neq \emptyset$ Observe that any continuous map, for which the set of one-point fibers is dense in the domain, is an irreducible map. If $f : X \to Y$ is any map with compact fibers, a Zorn's Lemma argument shows there exists a closed subset $Z \subseteq X$ such that $f \mid z : Z \to X$ is irreducible. N. Lašnev [9] showed that if $f : X \to Y$ is a closed surjection with X paracompact and Y Fréchet, then there exists closed $Z \subseteq X$ such that $f(Z) = Y$ and $f \mid z$ is irreducible. A closed surjection $f : X \to Y$ is said to be **inductively irreducible** if f has an irreducible restriction to some closed subspace of X. See [7] for results on when closed maps with a paracompact or Lindelöf domain are inductively irreducible.

In [12], J. van Mill and R.G. Woods discuss the perfect and perfect irreducible images of many of the classic zero-dimensional separable metric spaces. They characterize the perfect and perfect irreducible, zero-dimensional images of the Baire spaces $\mathbb{C}$ (***Cantor set***), $\mathbb{C} \setminus \{p\}$, and $\mathbb{P}$ (space of irrational numbers). One sample result: If $f : X \to Y$ is a perfect irreducible map from one of $\mathbb{C}$, $\mathbb{C} \setminus \{p\}$, or $\mathbb{P}$ there is a dense subset $S \subseteq Y$ such that $f|_{f^{-1}(S)} : f^{-1}(S) \to S$ is a homeomorphism, and, if Y is zero-dimensional, then Y is homeomorphic (respectively) to $\mathbb{C}$, $\mathbb{C} \setminus \{p\}$ or $\mathbb{P}$. They also prove that if Y is any σ-compact, nowhere locally compact metric space then there is a perfect irreducible map $f:\mathbb{Q} \times \mathbb{C} \to Y$ such that every fiber of f is homeomorphic to $\mathbb{C}$.

There are several classes of topological spaces which can be characterized as the class of images or preimages of another class under a closed or perfect map. One such class is the class of **Lašnev space**s [9] which is defined as the class of closed images of metric spaces. Any Laŝnev space Y, given by the image of a closed $f : X \to Y$ with X metrizable, can be "decomposed" as $Y = Y_0 \cup (\cup^{\infty}_{i=1} Y_i)$, where $f^{-1}(y)$ is compact for every $y \in Y_0$ and each Y_i, for $i \geqslant 1$, is closed discrete. Prior to Laŝnev's result K. Morita proved a similar decomposition theorem for closed images of paracompact locally compact spaces and since then the decomposition result has been extended to closed images of regular σ-spaces by J. Chaber and metacompact p-

Hart, K. P., et al. Encyclopedia of General Topology, Elsevier Science & Technology, 2003. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/willm/detail.action?docID=299054.
Created from willm on 2026-09-25 13:03:12.

Copyright © 2003. Elsevier Science & Technology. All rights reserved.

spaces by N.V. Veličko. See [HvM, Chapter 7] for a discussion of other extensions of Lašnev's theorem. L. Foged (see [HvM, Chapter 7]) obtained the following internal characterization of Lašnev spaces: A (Hausdorff) space X is a Lašnev space if and only if X is a Fréchet space with a σ-***hereditarily closure-preserving*** closed ***k-network***.

The class of paracompact p-spaces (equivalently, paracompact ***M-spaces***) can be characterized as the class of all perfect preimages of metric spaces [KV, Chapter 10]. Combining this with an earlier remark in this section about perfect maps we see that the class of paracompact p-spaces is exactly the class of all closed subsets of the product of a compact space with a metric space. The class of M-spaces is characterized as the class of quasi-perfect preimages of metric spaces. The class of paracompact Čech-complete spaces is characterized as the class of all perfect preimages of complete metric spaces [E, 5.5.9].

Every metric space of weight $\kappa \geqslant \#$ can be expressed as a perfect image of a subspace of the ***Baire space of weight*** κ, $B(\kappa)$ [E, 4.4.J], and every complete metric space of weight κ can be expressed as a closed image of $B(\kappa)$.

Any perfect image of the ordinal space ω_1 is just homeomorphic to ω_1 [14]. See [14] for a characterization of the quotients (perfect images) of compact ordinals. The perfect preimages of ω_1 can be much different than ω_1 and do not have to contain a copy of ω_1 as a subspace. For a quick example, let D be a discrete space of cardinality $\#_1$ and let $g : D \to \omega_1$ be a bijection. If $G : \beta D \to \omega_1 + 1$ is the Stone–Čech continuous extension of g then $Z = G^{-1}(\omega_1)$ is a perfect preimage of ω_1. Z cannot contain a copy of ω_1 since $\boldsymbol{Z}$ does not contain any nontrivial convergent sequences. D. Burke and G. Gruenhage (see [3] for reference) have independently shown that any noncompact space X, in which all countable sets have compact closure, contains a perfect preimage of ω_1. This can be useful in questions of how countably compactness relates to compactness. Assuming MA + ¬CH, this can be used to show that if X is a regular ***L-space*** and bX is any compactification of X then the remainder b$X \setminus X$ contains a perfect preimage of ω_1.

A continuous surjection $f : X \to Y$ is a **light map** (**zero-dimensional map**) if all fibers of f are ***hereditarily disconnected*** (respectively, zero-dimensional). A **monotone map** has all of its fibers connected. Zero-dimensional maps and monotone maps seem to be "opposite" notions. See [E, 6.2.22] for the result: If $f : X \to Y$ is a perfect map then f can be expressed as a composition $f = h \hat{x} g$ where $g : X \to Z$ is a monotone perfect map and $h : Z \to Y$ is a zero-dimensional perfect map.

If X and Y are metric spaces, a perfect map $f : X \to Y$ is said to be a **cell-like map** if the fibers $f^{-1}(y)$ have ***trivial shape*** for every $y \in Y$. Cell-like maps play a prominent role in geometric and infinite dimensional topology, particularly in the study of the ***Hilbert cube*** and Hilbert cube manifolds. With a "nice" domain and range, cell-like maps often turn out to be ***near-homeomorphisms***. See [11] for an introduction to cell-like maps.

It is useful to have a few examples of perfect or closed maps in order to see the types of possible constructions. In [2] an example is given of a perfect map from a locally compact space with a G_δ-diagonal onto a space with no G_δ-diagonal. The fibers in this map each have cardinality $\leqslant 2$. In [3] an example, due to Lutzer, is described of a Lindelöf semi-metrizable space with a countable ***network*** and a perfect map onto a space Y which is not of point-countable type and not symmetrizable, hence not first-countable or semi-metrizable. The perfect map in this case has just one fiber which is not a singleton. J. Chaber [6] has given an example of a perfect map from a p-space with a σ-locally countable base and a σ-disjoint base onto a space

Hart, K. P., et al. Encyclopedia of General Topology, Elsevier Science & Technology, 2003. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/willm/detail.action?docID=299054.
Created from willm on 2026-09-25 13:03:12.

Copyright © 2003. Elsevier Science & Technology. All rights reserved.

which is not a p-space. The construction of the domain space and resulting perfect map is general enough that it may have applications in other situations. In [3] there is an example, due to P. Zenor, of a locally compact, countably paracompact space and a closed map onto a space which is not countably paracompact. See [KV, Chapter 9] for an example by Burke of a perfect map from an orthocompact space onto a space which is not orthocompact. The fibers in this map are infinite – it is not known if orthocompactness is preserved under finite-to-one closed maps. See [15] for references to three examples which destroy submetrizability under finite-to-one perfect maps. These examples actually kill one of the properties of realcompactness, G_δ-diagonal or property wD. In [13] there is an open perfect map $f : X \to Y$ where is X is hereditarily disconnected (no nontrivial connected subsets) and Y is connected with a ***dispersion point***. Both X and Y can be embedded into $\mathbb{R}^3$.

References

[1] Aarts, J.M., Lutzer, D.J. Completeness properties designed for recognizing Baire spaces. *Dissertationes Math.*. 116, 1974.

[2] Burke, D.K. A nondevelopable locally compact Hausdorff space with a G_δ-diagonal. *Gen. Topology Appl.*. 1972;2:287–291.

[3] Burke, D.K. Closed mappings. In: Reed G.M., ed. *Surveys in General Topology*. New York: Academic Press, 1980.

[4] Burke, D.K., Hansell, R.W. Perfect maps and relatively discrete collections. In: *Papers on General Topology and Applications, Amsterdam, 1994*. New York: New York Acad. Sci.; 1996:54–56.

[5] Debs, G., Saint Raymond, J. Cofinal Σ_1^1 and Π_1^1 subsets of ω^ω. *Fund. Math.*. 1999;159:161–193.

[6] Chaber, J. Perfect images of p-spaces. *Proc. Amer. Math. Soc.*. 1982;85(4):609–614.

[7] 1–3 Gruenhage, G., Irreducible restrictions of closed mappings. 8th Prague Topological Symposium on General Topology and its Relations to Modern Analysis and Algebra. 1996. Topology Appl., 85. 1998:127–135.

[8] Hansell, R.W., Pan, S. Perfect images of Čechanalytic spaces. *Proc. Amer. Math. Soc.*. 1995;123(1):293–298.

[9] Lašnev, N. Continuous decompositions and closed mappings of metric spaces. *Soviet Math. Dokl.*. 1965;6: 1504–1506.

[10] Michael, E.A. A note on closed maps and compact sets. *Israel J. Math.*. 1964;2:173–176.

[11] van Mill, J., Infinite Dimensional Topology. North-Holland Math. Library, Vol. 43. Amsterdam: North-Holland, 1989.

[12] van Mill, J., Woods, R.G. Perfect images of zerodimensional separable metric spaces. *Canad. Math. Bull.* . 1982;25(1):41–47.

[13] Pol, R., Pol, E. Anopen-perfect mapping of a hereditarily disconnected space onto a connected space. *Fund. Math.*. 1975;86(3):271–278.

[14] Rajagopalan, M., Soundararajan, T., Jakel, D. On perfect images of ordinals. *Topology Appl.*. 1980;11(3): 305–318.

[15] Vaughan, J.E. Discrete sequences of points. *Topology Proc.*. 1978;3(1):237–266.

Hart, K. P., et al. Encyclopedia of General Topology, Elsevier Science & Technology, 2003. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/willm/detail.action?docID=299054.
Created from willm on 2026-09-25 13:03:12.